AI-Orchestrated Diagnosis and Repair of a VQE Ansatz Failure Mode

A real, reproducible worked example of using an AI agent (Claude, via Claude Code) to orchestrate a multi-week, compute-heavy quantum chemistry research workflow — submitted as the optional prototype for Track 1 (AI for Quantum Advantage on NISQ Hardware). Code and full results: `github.com/1shaanSr/vqe-ansatz-expressibility`

What this document is

This is not a description of how AI agents *could* be used for quantum research — it is an account of a specific project where one was, with the exact commands, failures, self-corrections, and validated results it produced. The underlying scientific result (Section 3) and the AI-orchestration process that produced it (Section 2) are both real and independently verifiable from the linked repository's commit history, run logs, and figures.

1. The research question

We study a chemistry-structured VQE ansatz (paired UCC doubles, PUCCD) across the potential energy surfaces of five molecules — H_2, LiH, BeH_2, HF, BH (4–12 qubits, STO-3G) — asking two questions: where does this ansatz fail, and is a given failure a fundamental limitation of what the ansatz can represent, or just an under-optimized run? The second question is rarely asked rigorously in small-scale VQE studies, and answering it is where AI-agent orchestration did the most work.

2. Why this fits Track 1

The project already includes a real IBM Quantum hardware run (backend `ibm_kingston`) using a single-batched-job strategy designed specifically to stay within free-tier QPU-time limits (all classical optimization performed in simulation; hardware used exactly once, for final verification). The proposed extension (Section 4) scales that same strategy to validate a newly-discovered ansatz fix on real hardware across full molecular curves — a direct, concrete push toward results that are harder to reach with today's NISQ devices under realistic access constraints.

2. The AI-orchestration workflow (the "how")

Three concrete episodes from this project illustrate what "AI-orchestrated" meant in practice — not passively running pre-written scripts, but actively managing long, failure-prone compute jobs and catching problems a human running the same pipeline could easily have missed.

2.1 Long-running background job orchestration

The full study spans 90+ bond-length geometries across 5 molecules, each requiring a multi-restart classical optimization (up to 12 restarts × 250 iterations per geometry). Individual runs took minutes to hours depending on system load, with highly variable per-point timing (observed range: ~150 seconds to over 2 hours for a single geometry under heavy system contention). The agent launched these as background processes, monitored them asynchronously without blocking on results, diagnosed and fixed real runtime bugs as they appeared (e.g. a silent NaN-target crash from 4 incomplete database records, a matplotlib API incompatibility), and re-launched corrected runs — all without requiring a human to babysit each step.

2.2 Catching its own overstated statistical claim

An early analysis reported a strong correlation between energy error and dipole-moment error (naive $p = 3.4\times10^{-5}$). Before accepting this, the agent identified that the underlying data points are *not* independent samples — they lie along one smooth curve — and applied an autocorrelation-corrected effective-sample-size test (Pyper & Peterman, 1998), which revised the result to a far less dramatic $p \approx 0.035$. A second self-audit, using outlier-robust Spearman rank correlation, further revealed that for two of the five molecules, an apparently strong raw correlation ($r = 0.75$–0.91) was being driven entirely by 1–2 extreme data points, not a real trend — and reported this honestly rather than keeping the more impressive-looking number.

2.3 Refusing to accept a good-looking result at face value

After a proposed ansatz fix showed a dramatic improvement (up to 446× error reduction) at 4 hand-picked, previously-catastrophic geometries, the agent did not stop there. It proactively re-ran the fix across the *full* set of 55 geometries for the affected molecules — precisely because a 4-point spot check could have been cherry-picked or masked a regression elsewhere on the curve. The full-curve run (Section 3) confirmed the fix generalized, and also surfaced an honest partial result (one molecule improved substantially but not completely) that a less rigorous process might have glossed over.

3. The scientific result this process produced (the "what")

The orchestration process above was in service of a real finding, not just activity for its own sake. Two distinct ansatz failure modes were identified and mechanistically explained, then one was resolved:

3.1 Two failure modes, explained by a classical (non-VQE) diagnostic

LiH's energy and dipole error degrade *gradually* as the bond stretches. HF and BH instead track the exact answer closely, then collapse sharply at one specific bond length. A purely classical diagnostic — the weight of the Hartree-Fock determinant in the exact (FCI) wavefunction, requiring no VQE run at all — explains why: HF and BH each undergo a genuine electronic-structure curve crossing at essentially the bond length where PUCCD fails; LiH does not.

3.2 A validated architectural fix

PUCCD is restricted to paired double excitations and has no orbital-rotation degree of freedom — exactly the capability needed to track a wavefunction across a crossing. This is grounded directly in Anselmetti et al. (*New J. Phys.* 23, 113010, 2021), who show that pair-exchange-only and orbital-rotation-only ansatze are each individually non-universal, while their combination is. Adding single (orbital-rotation) excitations to PUCCD was tested across all affected geometries, not just the original failure points:

Molecule	Geometries tested	Reached chemical accuracy	Max error reduction
HF	17	16 / 17 (was 0 / 17)	~38,000x
LiH	20	20 / 20 (was 0 / 20)	~29,000x
BH	18	1 / 18, every point improved	446x

Chemical accuracy threshold: 1.6 mHa. BH's partial result (reported honestly, not rounded up) is itself informative — it suggests boron's valence configuration retains residual multi-reference character that orbital rotation alone does not fully capture, a concrete lead for the proposed extension below.

3.3 Ruling out the obvious alternative explanation

Before attributing the original failures to the ansatz, insufficient optimization was explicitly ruled out: at the worst-affected geometries, 12 independent restarts (random and warm-started) converged to the identical energy to within 1×10^{-6} Ha regardless of starting point — a genuine expressibility ceiling, not an under-optimized run.

4. Proposed extension using Quantum Flywheel resources

The fix above is validated only in noiseless simulation. The proposed 3-month extension uses exactly the AI-orchestration pattern demonstrated in Section 2, now applied with real QPU access, more compute, and AI-agent-driven search:

- **Month 1 — QPU access.** Validate the orbital-rotation-augmented ansatz on real IBM hardware across full molecular curves, using the same single-batched-job strategy already proven in this project's prior hardware run, to stay within QPU-time budget.
- **Month 2 — cloud compute.** Extend to additional molecules and larger active spaces to test how far the fix generalizes, and investigate BH's residual gap (Section 3.2) — testing whether a complementary correction (e.g. a neural-network amplitude correction in the style of Li et al., 2025) closes it.
- **Month 3 — AI-agent-orchestrated search.** Rather than hand-selecting which additional excitation operators to test, use AI-agent-driven search over the remaining excitation space — proposing candidate ansatz modifications, running them, and applying the same self-auditing statistical methodology demonstrated in Section 2.2 to validate (not just report) each candidate result.

5. Why this is a low-execution-risk proposal

This is not a cold-start proposal. The pipeline, the hardware-execution strategy, the statistical methodology, and the diagnostic tooling all already exist and are already validated — the 3-month plan extends a demonstrated, working process rather than building one from scratch. All code, run scripts, raw results, and figures referenced in this document are public: `github.com/1shaanSr/vqe-ansatz-expressibility`.